# Burn ICU Case Study: Marcus Delaney, a Three-Day ICU Stay, Shift by Shift - FACULTY KEY

**Passing score: 78 of 100 points.**

## How to score (applies to every item)

- Score only what is written in the student's box. Spelling and grammar are not scored. Accept equivalent wording and correct synonyms. Give each element once.
- Partial points: use the amounts printed beside each element. Round the item total to the nearest 0.5. An item never earns more than its maximum.
- If a box is illegible or blank, give 0 and list the item on the score sheet under "faculty review."
- Chart answers are judged against the chart at the time written in the question. Small differences in totals read from the EHR (plus or minus 50 mL) are acceptable.

## About this patient

- Marcus Delaney, 34, 82 kg, 180 cm. House fire at 0130 Monday. Flame burn 22% TBSA (right arm circumferential, anterior chest, face and neck; about 8% full thickness) with smoke inhalation (COHb 14.2%), intubated at 0250, right forearm/hand escharotomy at 0315, Burn ICU arrival 0400.
- Parkland: 4 mL x 82 kg x 22% = 7,216 mL lactated Ringer's in 24 hours from the time of injury. Day 2 (Tuesday) he develops burn wound sepsis (MSSA), needs norepinephrine, and improves overnight; he is extubated Wednesday 1045 and transfers Thursday 0700.
- Shifts 1 and 2 have been written so far. The point total of the finished packet will be 100: Shift 1 = 20 (it includes the admission picture and the handoff), Shifts 2 to 6 = 16 each.

# PART 1: Shift 1: Monday 11/02, 0700 to 1900 (day shift) (20 pts)

*Shift 1 is mostly chart reading and the safety reasoning behind burn resuscitation. A careful student should earn about 16 of the 20 points. The hardest items are 1.2 (the Parkland math with the EMS bolus) and 1.10 (the fluid totals).*

## 1.1 Why Is He Here? Your Picture of Marcus at 0700 *[2.5 pts]*

*Set the clock to 0700. You are taking report from the night nurse. Read the ED Provider Note, the Burn Surgery and Critical Care H&Ps, the nursing notes from 0425 to 0650 and the Patient Summary. In your own words, explain what happened to Marcus, what injuries he has and what has already been done, then name the THREE most urgent problems you must manage over the next 12 hours.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Mechanism and burn: house fire about 0130 Mon, flame burns 22% TBSA (right arm circumferential, anterior chest, face/neck), about 8% full thickness (right forearm and hand). | ED note, Burn H&P | 0.5 |
| Inhalation injury and carbon monoxide: hoarse voice, stridor, soot, singed nasal hairs; COHb 14.2% at 0250; intubated 0250; bronchoscopy shows grade 2 inhalation injury. | ED note, bronchoscopy note | 0.5 |
| Right forearm/hand escharotomy at 0315 (tense forearm, no Doppler signal) with pulses returned. | Escharotomy note | 0.5 |
| Burn shock resuscitation: Parkland lactated Ringer's running at 450 mL/h and titrated to urine output; lactate 5.8 down to 2.2 by 0800 (3.4 at 0415). | Orders, labs | 0.5 |
| Ventilated and sedated (propofol, fentanyl); lines/tubes: left IJ central line, left radial arterial line, OG tube, Foley with temperature probe. | Flowsheets | 0.25 |
| Three priorities that make sense, for example: (1) airway/ventilation and carbon monoxide; (2) fluid resuscitation titrated to urine output; (3) circulation to the right hand after escharotomy (also acceptable: pain/sedation, infection prevention). | | 0.25 |

## 1.2 The Parkland Formula *[2 pts]*

*Find the burn size and his weight in the chart. (a) Calculate his total 24-hour fluid need with the Parkland formula. (b) How much must he receive in the FIRST 8 hours, and by what clock time does that period end? (c) EMS gave 500 mL of lactated Ringer's before the ED. Explain how that changes the hourly rate for the first 8 hours, counting from his ED arrival at 0230. (d) What rate is the second half of the fluid given at? Show your work with the units. Why is the clock started at the time of the injury and not at his arrival?*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Total: 4 mL x 82 kg x 22 = 7,216 mL lactated Ringer's in 24 hours (0.25 formula, 0.25 answer). | | 0.5 |
| First half = 3,608 mL in the first 8 hours from the INJURY (0130 to 0930). | | 0.5 |
| Subtract the EMS 500 mL: 3,108 mL over the 7 hours from 0230 to 0930 = about 444 mL/h (accept 450 mL/h, which matches the order). | Order: LR 450 mL/h | 0.5 |
| Second half = 3,608 mL over the next 16 hours (0930 to 0130 Tue) = 225 mL/h. | | 0.25 |
| The clock starts at the injury because the capillary leak and fluid loss begin then; the first hours are the most critical (any equivalent reason). | | 0.25 |

## 1.3 Vital Sign Trend *[1.5 pts]*

*Copy the vital signs at each time (he has an arterial line, so use the blood pressure shown). Circle every abnormal value. Then, in one or two sentences, describe the trend over the shift and say which value on the flowsheet you do NOT fully trust and why.*

| Vital | Mon 0700 | Mon 1100 | Mon 1500 | Mon 1900 |
|---|---|---|---|---|
| Temperature | 98.2 °F | 98.9 °F | 99.1 °F | 99.2 °F |
| Heart rate | 111 | 104 | 102 | 97 |
| Blood pressure | 99/57 | 105/62 | 107/62 | 109/63 |
| Respiratory rate | 18 | 18 | 17 | 17 |
| SpO2 (%) | 99 | 99 | 99 | 98 |
| Pain (CPOT) | CPOT 1/8 | CPOT 1/8 | CPOT 1/8 | CPOT 1/8 |

*Score 1.0 x (cells correct / 24), rounded to the nearest 0.5. The trend sentence is worth 0.5: heart rate starts high (111) and falls to 97 as the resuscitation works; blood pressure is low-normal (99/57 to 109/63, MAP about 71 to 78); temperature rises from 98.2 F*

*to 99.2 F (he arrived cold at 96.8 F); CPOT stays 1 because he is sedated. The value NOT to trust fully is the SpO2 of 99-100%: carbon monoxide on the hemoglobin makes the pulse oximeter read falsely normal. Allow +/- 2 for rates and +/- 0.2 F.*

### 1.3b Trend (scored with 1.3) *[0 pts]*

*Describe the trend over the shift and name the value you do NOT fully trust and why.*

Model answer: Scored with item 1.3.

### 1.4 The Pulse Oximeter Says 100% *[1.5 pts]*

*Marcus's SpO2 reads 99 to 100% all morning. (a) Explain why this number can be falsely reassuring in this patient. (b) Which lab results show how much carbon monoxide is on his hemoglobin, and what were the values at 0250, 0415 and 0800? (c) What treatment lowers it, and what did the night team change at 0600 (read their notes)? (d) He smokes a pack a day: what does that tell you about the value he is heading toward?*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| A pulse oximeter cannot tell carboxyhemoglobin from oxyhemoglobin, so it reads normal even though the blood carries much less oxygen. | | 0.5 |
| Carboxyhemoglobin (co-oximetry on the arterial blood gas): 14.2% at 0250, 6.4% at 0415, 3.1% at 0800. | Lab Results | 0.25 |
| 100% oxygen (FiO2 1.0 through the ventilator) shortens the half-life of carbon monoxide; FiO2 was lowered to 0.60 at 0600 once the level was below about 5%. | Flowsheets, MAR/orders | 0.5 |
| A smoker normally has a baseline COHb of about 3 to 5%, so a level around 3% is his baseline and not further poisoning. | | 0.25 |

### 1.5 Titrating the Fluids *[1.5 pts]*

*Open the lactated Ringer's order, the MAR and the hourly urine output. (a) What is the urine output goal in mL/h and in mL/kg/h? (b) At 0930 the rate was changed: from what to what, and why? (c) Write the urine outputs for 1200-1300 and 1300-1400, say what the order tells you to do, and what the new rate becomes at 1400. (d) Name two things you check when you change the rate and one danger of giving too much fluid.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Goal 30 to 50 mL/h, which is 0.5 mL/kg/h (about 41 mL/h for 82 kg). | Order instructions | 0.25 |
| At 0930 (8 hours after the injury) the first half of the Parkland fluid is done, so the rate dropped from 540 to 225 mL/h for the second 16 hours. | MAR | 0.25 |
| Urine output 28 mL and 28 mL: below 30 mL/h for two hours in a row, so increase the rate by 20 to 25% and notify the burn team. | I&O, order | 0.5 |
| 225 mL/h x 1.2 = 270 mL/h at 1400. | MAR | 0.25 |
| Two checks (lung sounds / peak airway pressures, abdomen or bladder pressure, right hand pulses and tension, edema, blood pressure, heart rate) AND a danger of too much fluid ("fluid creep": pulmonary edema, compartment syndrome, more ventilator days, worse swelling). | | 0.25 |

### 1.6 Calling the Burn Surgeon *[1.5 pts]*

*It is 1400 and you have just seen the second low urine output. Write the SBAR you give Dr. Okafor on the phone. Use real numbers from the chart. End with what you are asking him to do and how you will confirm the order.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Situation: identify yourself, the patient and room; urine output 28 mL/h and 28 mL/h for two hours (about 0.34 mL/kg/h, less than the goal of 0.5) on lactated Ringer's 225 mL/h. | | 0.375 |
| Background: 34-year-old, 82 kg, 22% TBSA flame burn with inhalation injury, intubated and sedated, on the Parkland resuscitation since 0230, EMS 500 mL, escharotomy right arm. | | 0.375 |
| Assessment: current vitals (T 99.0, HR 103, BP 105/61, SpO2 99%), lungs/peak pressures, right hand Doppler pulses, lactate normal at 1.8 (1200), CK 1,120, and your concern that he is behind on fluid. | | 0.375 |
| Recommendation: ask to increase the rate by 20% per the order (to 270 mL/h) and what urine output you should call about next; read the order back and document the time and name. | | 0.375 |

## 1.7 Lines, Tubes and Airway *[1.5 pts]*

*List SIX lines, tubes or devices Marcus has at 0700 (not counting the ventilator). For each, write the site and what you check every shift, and one complication you are watching for.*

Model answer: Each device row is worth 0.25: the device and site are right AND the check/complication fit that device.

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Endotracheal tube, 8.0 oral, 24 cm at the teeth: confirm depth and securement, cuff pressure (about 24), HOB 30, oral care, suction as needed. Complications: accidental extubation, VAP, pressure injury to the lips, tube obstruction. | | 0.25 |
| Left internal jugular central line (triple lumen): chlorhexidine dressing intact, scrub the hub, daily need review, label lumens. Complications: infection (CLABSI), air embolism, clot, bleeding. | | 0.25 |
| Left radial arterial line (right arm is burned): waveform, level and zero, check the left hand circulation, pressure bag 300 mmHg. Complications: hand ischemia, bleeding or disconnection, infection. | | 0.25 |
| OG tube 16 Fr: position confirmed by x-ray, mark at the lips, residual and feeds per order, HOB 30. Complications: aspiration, misplacement, intolerance (residual, distention). | | 0.25 |
| Foley catheter 16 Fr with temperature probe: hourly output, secured, closed system, catheter care. Complications: CAUTI, kinks/low output that is really a blocked catheter, bladder pressure reading for compartment syndrome. | | 0.25 |
| Left antecubital 18 g peripheral IV: site check, flush, patency. Complications: infiltration, phlebitis, infection. (Also accept the escharotomy sites on the right arm as a device row.) | | 0.25 |

## 1.8 The Ventilator and Sedation *[1.5 pts]*

*Open Flowsheets > Respiratory at 0700. (a) Calculate his ideal body weight (male: 50 kg + 2.3 kg for every inch over 5 feet; 1 inch = 2.54 cm) and the tidal volume at 6 mL/kg. Does this match the set tidal volume? (b) List FOUR actions in the ventilator-associated pneumonia (VAP) bundle that you do this shift. (c) What is the RASS goal on Monday, and why is he sedated more deeply on the first day?*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| 180 cm = 70.9 inches, so IBW = 50 + 2.3 x 10.9 = about 75 kg; 6 mL/kg = about 450 mL. The set tidal volume is 450 mL, which matches (accept 75-76 kg and 450-455 mL). | | 0.75 |
| Any four VAP bundle actions: head of bed 30 degrees or higher, oral care with chlorhexidine every 12 h, daily sedation interruption and spontaneous breathing trial, cuff pressure check, suction only when needed, stress ulcer and DVT prophylaxis, hand hygiene (0.125 each). | | 0.5 |
| RASS goal -2 to -3 on Monday; deeper sedation and analgesia are needed on day 1 for painful burn care, resuscitation, the airway swelling and the risk of pulling the tube. | | 0.25 |

## 1.9 Reading the Labs *[1.5 pts]*

*Open Lab Results and compare the first draws (about 0240-0250) with the 1200 and 1800 draws. Choose FOUR labs or lab groups that changed or matter. For each, write the values, why this happens in a burn patient, and what you do about it.*

Model answer: Any 4 of the labs below; each row 0.375: values correct (0.125), reason correct (0.125), nursing action reasonable (0.125).

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Lactate 5.8 -> 3.4 -> 2.2 -> 1.8 -> 1.6 (critical to normal): poor tissue perfusion early; falling means the fluid resuscitation is working. Keep trending; a rise means shock or sepsis, notify. | | 0.375 |
| Hemoglobin 17.2 -> 13.6 (and hematocrit 51 -> 41): hemoconcentration from plasma leak on arrival, then dilution as fluid is given. Watch for bleeding and new drops. | | 0.375 |
| CK 480 -> 1,120: muscle and tissue injury; protect the kidneys with adequate urine output and watch for dark urine (myoglobin); report the trend. | | 0.375 |
| Potassium 4.3 -> 4.5 -> 4.9 -> 5.0: cells release potassium after burn injury; monitor the ECG and recheck; do not replace it, and report a value of 5.1 or higher. | | 0.375 |
| WBC 12.8 -> 16.1 (1200): stress response to a major burn, NOT an infection yet; use it as a baseline for comparison later. | | 0.375 |
| Glucose 143 -> 156 -> 172 -> 188 (serum): stress hyperglycemia that keeps rising; check glucose | | 0.375 |

*Score only what is written. Flag anything you cannot read for faculty review.*

| | | |
|---|---|---|
| per order and know the call parameters, because an insulin infusion is likely if it stays above 180. | | |
| pH 7.29 -> 7.39 and bicarbonate 18 -> 24 (arterial gas): metabolic acidosis from poor perfusion corrected with resuscitation. | | 0.375 |
| Carboxyhemoglobin 14.2 -> 3.0 (also accept if not used in 1.4). | | 0.375 |

## 1.10 Adding Up Your 12 Hours *[1.5 pts]*

*Open Flowsheets > Intake/Output and add the hourly rows from 0700 to 1900 (12 rows). (a) What are his total intake, total output and net balance for your shift? (b) What was his average urine output in mL/h and in mL/kg/h? (c) Using the Parkland rates, how much lactated Ringer's should he have received BY 1900 counting from the injury (the EMS 500 mL and the ED 675 mL before the ICU count), and how much did he actually receive in all? (d) What does that tell you?*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Intake about 4,490 mL, output about 460 mL, net about +4,030 mL (accept totals within 50 mL; a large positive balance is expected in burn resuscitation). | I&O | 0.5 |
| Average urine output about 38 mL/h = about 0.47 mL/kg/h (just under the 0.5 goal, inside 30-50 mL/h most of the shift). | | 0.25 |
| Expected by 1900: first 8 h = 3,608 mL plus 9.5 h x 225 = about 2,140 mL, about 5,750 mL. Actual crystalloid: EMS 500 + ED 675 + ICU LR 5,153 = about 6,330 mL. | | 0.5 |
| He is about 580 mL (10%) over the Parkland estimate: acceptable because the order is to titrate to urine output, but a reminder to watch for fluid creep (edema, airway pressures, bladder pressure) and not to chase a single low urine output. | | 0.25 |

## 1.11 The 1400 Dressing Change *[1.5 pts]*

*Read the Burn Wound Assessment note and the 1520 nursing note. (a) Why was hydromorphone given at 1345 and what tool measures pain in an intubated, sedated patient (name its four parts)? (b) Which topical drugs were applied and where? (c) What do you assess at every dressing change? (d) Name two ways you protect him and yourself from infection.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Dressing changes are the most painful part of burn care; the opioid is given about 15 minutes BEFORE the procedure so it is working. CPOT has four parts: facial expression, body movements, muscle tension, compliance with the ventilator. | | 0.5 |
| Silver sulfadiazine 1% cream to the chest and right arm; bacitracin ointment to the face. | MAR | 0.25 |
| Wound color, depth, drainage amount and odor, eschar, blisters, edges and any redness beyond the burn margin (mark and time it), skin around the wound, right hand pulses and capillary refill, and pain response. | | 0.5 |
| Any two: sterile technique, gown/gloves/mask and contact precautions, hand hygiene, dedicated equipment, cleansing the old cream, covering the wound promptly. | | 0.25 |

## 1.12 Handoff at 1900 *[2 pts]*

*It is 1850. Write your SBAR handoff for E. Haddad, RN, the night nurse. Use real numbers and name what the night nurse must watch for and do in the first part of the night, including anything coming up in the next few hours.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Situation: patient, room, full code, 22% TBSA burn with inhalation injury, intubated and sedated (propofol and fentanyl), current vitals (HR about 97, BP 109/63, T 99.2, SpO2 98%), resuscitation going as planned. | | 0.5 |
| Background: house fire 0130, escharotomy right arm 0315, Parkland (LR 7,216 mL goal in 24 h), lines (central line, arterial line, OG with feeds started at 1000, Foley), COHb now at his smoker baseline. | | 0.5 |
| Assessment: urine output was 28 mL/h around 1200-1400 and corrected after the rate went to 270 mL/h (now 42-46 mL/h); net about +4 L; lactate 1.6; potassium 5.0 and glucose 188 are climbing; right hand Doppler pulses steady; pain controlled; wound care done at 1400. | | 0.5 |
| Recommendation / to do: hourly urine output with titration per order; hourly right hand checks; the Parkland 24 hours ends at 0130 Tuesday and the order changes then; 0000 labs (glucose may need an insulin infusion; recheck potassium); sedation goal and restraint checks; turning and skin care; family visiting; call the burn team for UOP under 20 mL/h. | | 0.5 |

*Score only what is written. Flag anything you cannot read for faculty review.*

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# PART 2: Shift 2: Monday 11/02 1900 to Tuesday 11/03 0700 (night shift) (16 pts)

*Shift 2 tests the "quiet" overnight judgment: the end of the Parkland period (1.2 math repeated with the final total), the insulin infusion, an agitation event in a ventilated patient with an alcohol history, and compartment checks. A careful student should earn about 12 of the 16 points.*

## 2.1 Vital Sign Trend Through the Night *[1.5 pts]*

*Set the clock to 1900 on Monday. You take report from the day nurse. Copy the vital signs at each time, circle every abnormal value, then write two sentences below: what the night looked like overall, and what the flowsheet does NOT show about the agitation event that happened around 0245.*

| Vital | Mon 1900 | Mon 2300 | Tue 0300 | Tue 0700 |
|---|---|---|---|---|
| Temperature | 99.2 °F | 99.3 °F | 99.3 °F | 99.4 °F |
| Heart rate | 97 | 97 | 93 | 96 |
| Blood pressure | 109/63 | 109/62 | 109/64 | 111/64 |
| Respiratory rate | 17 | 16 | 16 | 17 |
| SpO2 (%) | 98 | 99 | 98 | 98 |
| Pain (CPOT) | CPOT 1/8 | CPOT 1/8 | CPOT 1/8 | CPOT 1/8 |

*Score 1.0 x (cells correct / 24), rounded to the nearest 0.5. The two sentences are worth 0.5: (1) a quiet night: heart rate 93-97 (borderline-high at 97, better than yesterday), BP about 109-111/62-64 (MAP about 78), temperature 99.2-99.4 F, SpO2 98-99%, CPOT 1; (2) the hourly flowsheet does not show the 0245 agitation: the vital signs at 0300 (HR 93, BP 109/64, RASS back to -3 after lorazepam) look normal, so an event can happen between hourly sets and only the nursing note and the RASS record it. Allow +/- 2 for rates and +/- 0.2 F.*

## 2.1b Overall night, and what the vitals do not show (scored with 2.1) *[0 pts]*

Model answer: Scored with item 2.1.

## 2.2 Glucose and the Insulin Infusion *[2 pts]*

*Open Lab Results (Glucose, point of care and serum), Orders, the MAR and the 0150-0320 nursing notes. (a) List his glucose values from 1800 through 0300 and explain why the insulin infusion was started at 0030 and at what rate. (b) Why does a burn patient have a high glucose even though he has no diabetes? (c) What is the glucose goal and how often do you check it on the infusion? (d) Give TWO safety actions for an insulin infusion, including what you must do if the tube feeding is stopped, and name one lab that you watch closely with insulin.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Glucose: 188 (1800), 198 (2200), 196 (0000), then 191, 190, 189 on the infusion. Started at 1 unit/h at 0030 because the glucose was above 180 on consecutive checks (ICU protocol), ordered by Dr. Brandt. | Labs, notes | 0.5 |
| Stress response of a major burn: stress hormones (cortisol, catecholamines, glucagon) and inflammation make the body insulin resistant and the liver release glucose; the hypermetabolic state and the tube feeding add to it. | | 0.5 |
| Goal 140 to 180 mg/dL; glucose every hour while on the infusion, and adjust the rate per the protocol table. | Order | 0.5 |
| Any two safety actions: independent double check of the insulin bag and rate (high-alert drug); do not change the rate without the protocol; if feeds are stopped or interrupted, call the provider and recheck the glucose because hypoglycemia is likely (treat glucose under 70 per protocol); dedicated line/label; check glucose after any rate change. AND one lab watched: potassium (insulin moves potassium into cells; K 4.5 -> 3.9 overnight). | | 0.5 |

## 2.3 The End of the First 24 Hours *[1.5 pts]*

*The Parkland period ends at 0130 (24 hours after the injury). Using the lactated Ringer's rates and times on the MAR (EMS 500 mL, ED 450 mL/h from 0230 to 0400, then the ICU rates), (a) add up the total crystalloid he received from the injury to 0130 and compare it with the Parkland goal. (b) What did the order change to at 0130 and what new fluid starts at 0600? (c) Why do the fluids not simply stop at 24 hours? (d) His weight was 82 kg on arrival and is now 89.6 kg: what does that tell you and what else do you watch for?*

*Score only what is written. Flag anything you cannot read for faculty review.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Total: 500 (EMS) + 675 (ED 0230-0400) + 900 (0400-0600 at 450) + 1,890 (0600-0930 at 540) + 1,012 (0930-1400 at 225) + 3,105 (1400-0130 at 270) = about 8,080 mL, about 864 mL (12%) over the 7,216 mL goal (accept 8,000-8,150). | MAR | 0.5 |
| At 0130 lactated Ringer's was lowered to a maintenance rate of 125 mL/h; albumin 5% 500 mL over 4 hours starts at 0600 per the burn protocol. | Orders | 0.375 |
| The capillary leak and losses continue for 24 to 48 hours, so he still needs fluid; colloid (albumin) holds fluid in the vessels and lets you use less crystalloid. | | 0.25 |
| Weight is up 7.6 kg, so the extra fluid is in the tissues (edema); also watch lung sounds and ventilator pressures, bladder pressure, the right arm compartments and urine output. | | 0.375 |

## 2.4 Alcohol and the Next Three Days *[1 pts]*

*His ethanol level was 42 mg/dL at 0240 and he usually drinks 4 to 6 beers on weekends, more than that the night of the fire. (a) When could alcohol withdrawal begin, and when are seizures and delirium tremens most likely? (b) Why may you not see the signs yet? (c) Which two vitamins are ordered and why? (d) Name three signs you would watch for.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Withdrawal can begin 6 to 24 hours after the last drink; seizures 12 to 48 hours; delirium tremens 48 to 96 hours, so the highest risk for him is Wednesday to Thursday. | | 0.25 |
| Propofol, fentanyl and lorazepam are sedating and mask the signs; and he cannot talk. | | 0.25 |
| Thiamine (prevents Wernicke encephalopathy, needed before or with glucose) and folic acid (deficiency in alcohol use). | MAR | 0.25 |
| Any three: tachycardia, hypertension, fever, tremor, sweating, agitation, hallucinations, seizure, difficulty sedating (needing more drug); use RASS and CAM-ICU because CIWA needs a patient who can talk. | | 0.25 |

## 2.5 The Agitation Event at 0245 *[2 pts]*

*Read the 0320 nursing note. Marcus (RASS -3 earlier) becomes restless, thrashes his head, bites the endotracheal tube and the ventilator pressure alarms sound. (a) In order, what do you check and do FIRST, before any medication, and why? (b) The provider orders lorazepam 1 mg IV. Name one concern about giving a benzodiazepine to this patient and one question you could ask the provider. (c) What are the rules for the soft wrist restraint you are using? (d) Besides the 0300 recheck, what do you document and who do you notify?*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Stay with him and check for causes before sedating: airway/tube (depth 24 cm, cuff pressure, secretions, kinks, bite block), ventilator alarms and synchrony (call RT), oxygenation, pain (CPOT 1-2), full bladder or Foley problem, positioning, and withdrawal/delirium; keep him safe (no unplanned extubation). Give analgesia first if pain is possible. | Nursing note | 0.75 |
| Concern: benzodiazepines increase delirium, prolong ventilation and can mask withdrawal; the unit prefers analgesia-first and propofol/dexmedetomidine. A good question: can we increase the propofol or fentanyl instead, or is a one-time lorazepam dose acceptable while he is ventilated (reason: alcohol-withdrawal risk)? Lorazepam is high-alert: double check the dose. | | 0.5 |
| Restraint rules: provider order that is renewed every calendar day, least restrictive method, check skin/circulation and release at least every 2 hours, assess the continuing need, document, remove at extubation, family informed. | Order | 0.5 |
| Document the event (time, behavior, cause search, response, RASS/CPOT before and after, lorazepam 1 mg IV at 0300 with the reason), notify Dr. Brandt (SBAR) and the charge nurse, and update the family when they call. | | 0.25 |

## 2.6 Checking the Right Hand and the Abdomen *[1.5 pts]*

*Compartment syndrome is a risk after a circumferential burn and a large fluid resuscitation. For each of the six Ps, write what you can or cannot check in Marcus (he is sedated and intubated) and what the chart showed at 2300-0300. Then name the abdominal check ordered every six hours.*

Model answer: Rows 1-6 worth 0.2 each (the P is named, a sensible way to check, and the chart finding); row 7 (bladder / intra-abdominal pressure) 0.3.

*Score only what is written. Flag anything you cannot read for faculty review.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Pain: out of proportion, with passive stretch; cannot be reported while sedated, so look for grimace/CPOT rise and a tense forearm. Chart: CPOT 1. | | 0.2 |
| Pallor (and cool temperature): fingers pink and warm. Chart: warm, cap refill 3 seconds. | | 0.2 |
| Pulselessness: Doppler signals over the radial and ulnar arteries (cannot rely on palpation through edema). Chart: biphasic Doppler signals present. | | 0.2 |
| Paresthesia: cannot be asked; sensation is not assessable while sedated. Chart: not testable. | | 0.2 |
| Paralysis: motion of the fingers, only when sedation is lightened (SAT at 0800). Chart: not testable yet. | | 0.2 |
| Poikilothermia / tension: skin temperature compared with the left hand, and firmness of the forearm compartments. Chart: forearm soft and compressible. | | 0.2 |
| Bladder (intra-abdominal) pressure every 6 hours through the Foley: 12 mmHg at 2330; call the provider above 15 (abdominal compartment syndrome is above 20 with organ dysfunction). | Order, note | 0.3 |

## 2.7 Tube Feeding and Calories *[1.5 pts]*

*Open the tube feeding order and the flowsheet. (a) Write the advancing schedule from 1000 Monday and the rate at 0600 on Tuesday. (b) The formula is Osmolite 1.5, which has 1.5 kcal per mL and 62.7 g of protein per liter: how many calories and grams of protein will he receive in 24 hours at 70 mL/h? (c) Give two reasons burn patients are fed early through the gut. (d) What do you check before advancing, and what do you do for a residual over 500 mL?*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Start 20 mL/h at 1000, then increase 10 mL/h every 4 hours (30 at 1400, 40 at 1800, 50 at 2200, 60 at 0200) to the goal of 70 mL/h at 0600. | Order | 0.25 |
| 70 mL/h x 24 h = 1,680 mL; x 1.5 kcal/mL = 2,520 kcal; x 62.7 g/L = about 105 g protein (accept 100-110). | | 0.5 |
| Any two: burns make the body hypermetabolic (needs about 2,400-2,500 kcal and extra protein); early feeding protects the gut lining and lowers bacterial translocation and infection; supports wound healing; reduces muscle loss and stress ulcers. | | 0.5 |
| Check the position of the tube (OG at the mark), the residual, abdominal distention, bowel sounds, stool, glucose, and keep the head of the bed at 30 degrees; for a residual over 500 mL hold the feeding and call the provider (and do not hold feeds for dressing changes). | | 0.25 |

## 2.8 The Midnight and 0600 Labs *[1.5 pts]*

*Open Lab Results and compare the 0000 and 0600 draws (and the 0500 albumin and coagulation labs) with the day shift values. Choose FOUR results. Write the values, why they happen in a burn patient on propofol and tube feeding, and what you do or ask for.*

Model answer: Any 4 rows of the bank; each row 0.375: values correct (0.125), reason (0.125), action (0.125).

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Phosphorus 3.4 -> 2.4 -> 2.1 (low): burn losses, insulin and feeding drive phosphate into cells (refeeding), dilution. Report to the provider; replacement is usually ordered; low phosphorus weakens breathing muscles and the heart. | | 0.375 |
| Magnesium 1.7 -> 1.6 (low): dilution and losses; report for replacement; check before the potassium. | | 0.375 |
| Potassium 4.5 -> 3.9: the insulin infusion and feeds move potassium into cells; keep it 4.0 or higher; notify the provider and recheck with the next draw. | | 0.375 |
| Albumin 2.6 (low) and prealbumin 7.1 (low): plasma leak, dilution and the acute-phase response; they do not measure nutrition well in the first days of a burn, but explain the albumin 5% order and the dietitian follow-up. | | 0.375 |
| Triglycerides 168 (high): propofol is a lipid emulsion (1.1 kcal/mL); report it, count the calories, and a level over 400 means ask to change the sedative. | | 0.375 |
| CK 1,120 -> 840 (falling): muscle injury is improving; keep adequate urine output. | | 0.375 |
| Hemoglobin 12.4 and platelets 214: dilution; platelets still normal; watch for bleeding with heparin prophylaxis. WBC 15.2 is stress response, a baseline for later comparison. | | 0.375 |

*Score only what is written. Flag anything you cannot read for faculty review.*

| | | |
|---|---|---|
| Sodium 134 (low): lactated Ringer's has less sodium than the plasma and he is retaining fluid; watch; no correction needed now. | | 0.375 |

## 2.9 Propofol: Math and Safety *[1.5 pts]*

*At 1900 on Monday the propofol is running at 35 mcg/kg/min. The bag is propofol 10 mg/mL and his weight is 82 kg. (a) Calculate the rate in mL/h. (b) Propofol gives 1.1 kcal per mL: how many calories does the sedation add in 24 hours? (c) Name two reasons you check the triglyceride level and two signs of propofol infusion syndrome. (d) Name two nursing actions for the bag and tubing.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| 35 mcg/kg/min x 82 kg = 2,870 mcg/min x 60 = 172,200 mcg/h = 172.2 mg/h; at 10 mg/mL = 17.2 mL/h. | | 0.5 |
| 17.2 mL/h x 24 h = about 413 mL/day x 1.1 kcal = about 455 kcal/day (accept 450-460); count it as part of his calories. | | 0.25 |
| Triglycerides: lipid load can raise them (his 168 is mildly high) and very high levels cause pancreatitis; check about every 2 to 3 days. Infusion syndrome: unexplained metabolic acidosis, rhabdomyolysis/rising CK, hyperkalemia, arrhythmia or heart failure, high triglycerides, kidney injury (any two). | | 0.5 |
| Any two: aseptic technique and change the tubing and vial/bag every 12 hours (lipid grows bacteria), label, watch blood pressure with every rate increase, do not mix other drugs in the line, daily sedation interruption. | | 0.25 |

## 2.10 Handoff at 0650 *[2 pts]*

*It is 0650 on Tuesday. Write your SBAR handoff for J. Morales, RN, the day nurse. Use real numbers, include the events of the night, what is happening in the next few hours (look at the orders and MAR), and anything you would like the day team to follow up.*

| Required element (accept equivalent wording) | Where / who | Pts |
|---|---|---|
| Situation: patient/room, full code, burn 22% with inhalation injury, intubated; stable night: HR 96, BP 111/64, T 99.4, SpO2 98%, RASS -3, CPOT 1; urine output 41-47 mL/h. | | 0.5 |
| Background: the first 24 hours of Parkland finished at 0130 (about 8,080 mL), LR now 125 mL/h (about to drop to 75 when the albumin starts at 0600); insulin infusion since 0030; feeds at goal 70 mL/h; weight 89.6 kg. | | 0.5 |
| Assessment: agitation at 0245 treated with lorazepam 1 mg (calm since); glucose 179-182 on the infusion (1.5 units/h from 0600); phosphorus 2.1, magnesium 1.6 and potassium 3.9 low, reported to Dr. Brandt, replacement not yet given; right hand signals steady; bladder pressure 12; triglycerides 168; skin and turns done. | | 0.5 |
| Recommendation / to do: sedation interruption at 0800; replace phosphorus, magnesium and potassium when ordered and recheck; hourly glucose; albumin 5% 500 mL running 0600-1000; dressing change and silver sulfadiazine at 1400 with hydromorphone beforehand; restraint order renewal; update Jenna and Linda; chest x-ray read; watch for alcohol withdrawal. | | 0.5 |

# SCORE SHEET

Student: _____ Date graded: _____ Passing score: 78 of 100

| Item | Description | Max | Earned |
|---|---|---|---|
| **Part 1** | **Shift 1: Monday 11/02, 0700 to 1900 (day shift)** | **20** | |
| 1.1 | Why Is He Here? Your Picture of Marcus at 0700 | 2.5 | |
| 1.2 | The Parkland Formula | 2 | |
| 1.3 | Vital Sign Trend | 1.5 | |
| 1.3b | Trend (scored with 1.3) | 0 | |
| 1.4 | The Pulse Oximeter Says 100% | 1.5 | |
| 1.5 | Titrating the Fluids | 1.5 | |
| 1.6 | Calling the Burn Surgeon | 1.5 | |
| 1.7 | Lines, Tubes and Airway | 1.5 | |
| 1.8 | The Ventilator and Sedation | 1.5 | |
| 1.9 | Reading the Labs | 1.5 | |
| 1.10 | Adding Up Your 12 Hours | 1.5 | |
| 1.11 | The 1400 Dressing Change | 1.5 | |
| 1.12 | Handoff at 1900 | 2 | |
| **Part 2** | **Shift 2: Monday 11/02 1900 to Tuesday 11/03 0700 (night shift)** | **16** | |
| 2.1 | Vital Sign Trend Through the Night | 1.5 | |
| 2.1b | Overall night, and what the vitals do not show (scored with 2.1) | 0 | |
| 2.2 | Glucose and the Insulin Infusion | 2 | |
| 2.3 | The End of the First 24 Hours | 1.5 | |
| 2.4 | Alcohol and the Next Three Days | 1 | |
| 2.5 | The Agitation Event at 0245 | 2 | |
| 2.6 | Checking the Right Hand and the Abdomen | 1.5 | |
| 2.7 | Tube Feeding and Calories | 1.5 | |
| 2.8 | The Midnight and 0600 Labs | 1.5 | |
| 2.9 | Propofol: Math and Safety | 1.5 | |
| 2.10 | Handoff at 0650 | 2 | |
| | **TOTAL** | **36** | |

Result: [ ] PASS (78 or more) [ ] DOES NOT PASS (below 78) Items needing faculty review (illegible or unclear): _____

*Score only what is written. Flag anything you cannot read for faculty review.*